

Original Article



Assessing the challenges in the referral system: a case study in the Hohoe Municipality

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Conflict of Interest

The authors declare that they have no competing interests.

ABSTRACT

Background: An effective referral system ensures a patient receives optimal health care at an appropriate health care level, specifically in times of medical emergency. A well-functioning referral system should allow the continuity of health care across the different tiers of care. Contrary, Ghana's health care referral system has not been given the needed attention despite the significant role an effective referral system can play in saving the lives of patients. This study explored the challenges of the referral system in the Hohoe Municipality.

Methods: This study used a phenomenological qualitative research approach. Health workers and relatives of patients who have been through the referral system were recruited for the study. In-depth interviews were conducted using an open-ended interview guide. The interviews were tape-recorded with permission from the participants who had been through the referral system and health workers in the Hohoe Municipality. Thematic analysis was used to analyze the data. Qualitative software package Atlas. ti was used in generating codes and themes.

Results: The results of the study show that inadequate human resources like medical experts and infrastructures like the unavailability of enough beds and medical supplies, were some of the challenges faced by the health workers. Socio-economic factors like not being able to afford the means to get to the next facility, also bad roads, unavailability of some medical supplies like oxygen were not readily available, bad attitude, and reluctance from some health workers were some of the challenges encountered by the relatives who have been through the referral system.

Conclusion: Referral is very important in the continuity of care and if the challenges are not being looked at more lives would be lost. Road networks, human infrastructures, and medical supplies are needed for the smooth running of the referral system.

Keywords: Ambulance; Referral; Patients; Relatives

INTRODUCTION

An effective referral system is vital for ensuring continuity and quality of healthcare, especially in decentralized health systems like Ghana's. It connects different levels of care

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community, primary, secondary, and tertiary so that patients receive the right services at the right time. In the Hohoe Municipality, the referral system plays a crucial role in managing health service delivery across Community-based Health Planning and Services (CHPS) compounds, health centers, and hospitals. However, challenges such as poor communication, inadequate transport, and weak feedback mechanisms often disrupt timely referrals and patient outcomes.^{1,2} Understanding how the referral system functions in Hohoe is therefore essential for improving coordination and achieving equitable access to healthcare services in Ghana. A good referral system can also help to ensure patients receive optimal care at the appropriate level. This reduces unnecessary charges to enhance the continuity of health care.³

In studies examining child mortality in several sub-Saharan African and South Asian settings, about 64% of deceased infants and children had at least one delay (in seeking care, transport, or facility care) before death.⁴ These deaths occur as a result of a delay in referring patients on time. Referral of patients from less equipped and skilled personnel and delay in reaching referred facilities on time due to poor road network coupled with poor means of transport also increases mortality.⁵ There are insufficient infrastructures to help run the referral system smoothly. Also, the distance to more equipped facilities mostly in developing countries contributes to the global burden of people who die annually as the result of poor referral systems.⁶

In rural Africa, referral between the first and second levels of health care systems is a major challenge.⁷ Some of these challenges faced oftentimes during the referral of patients include poor service quality, low availability of trained personnel, inadequate supplies of drugs and medical diagnostic equipment, and inadequate communication between the two facilities.⁵ Additionally, the largely disproportional distribution of skilled trained medical health professionals especially in the urban areas to the detriment of rural dwellers. This poses a greater health threat most importantly when on medical referral.^{7,8} Noted that Ghana's referral system from a lower facility to a higher facility is a major challenge. Most of the higher facilities which are well equipped with skilled personnel are concentrated within the southern part of the country.⁹ The geographical features such as mountains and valleys on the way make it unmotorable, especially during the rainy seasons, and poor network connectivity from the referral-to-referred facility is very challenging.

The unavailability of emergency services and transport in planning CHPS strategy makes it challenging and life-threatening for a Ghanaian patient on referral.¹⁰ Late referral could result in severe health complications and even result in death.¹¹ Even though the CHPS strategy has measures for patients in an emergency condition, oftentimes, these facilities have no available vehicle for immediate transfer of a patient in need of medical emergency services.¹² Studies focused on referrals in some parts of the country on various aspects of the referral system such as the quality of the referral letters, the communication challenges and feedback to the referral point, road network challenges, and the means of transport to the next point of care. However, none has looked at patients' and health personnel's perspectives on the challenges of the referral system. This study employed a qualitative research method to ascertain the necessary evidence from health authorities in the Hohoe Municipality. Specifically, the study explored the perspective of health workers' challenges faced during the referral, and the perspective of patients regarding the challenges they face during the referral. Although there are other publications on referral systems; however, our study explored the phenomenon from the patients' and care providers' perspectives. Giving actual experience to patients and caregivers regarding how the system works for them. This will

enhance the understanding of the referral system and the needed actions to strengthen the system, considering various perspectives.

METHODS

Study design

This study used a phenomenological qualitative study design approach. Face-to-face interviews were conducted using an interview guide (**Supplementary Data 1**). A phenomenological study focuses on the lived experiences of individuals within the world. This study explores the challenges patients, relatives, and health workers experience in the referral process. This helps the researcher to explore the challenges of the referral process.

Study setting

The study was conducted in the Hohoe Municipality in the Volta region of Ghana. Specifically, the Hohoe municipal hospital and Alavanyo Dzogbedze CHPS compound. The Municipality consists of 102 communities with a land area of 1.172 sq. km. The population of the Municipality is 167,016.¹³ The growth rate is 2.5%, representing 7.9% of the total Volta Region population. Out of this population, females represent 52.1%, and 47.9% are males. The Municipality shares common boundaries with Jasikan district to the north, Biakoye district to the north-west, Kpando Municipality by west and south-west, Afadjato district by the south, and the Republic of Togo by the east. The Hohoe Municipality has a total of 26 healthcare facilities. These facilities include a hospital, a polyclinic, a health center, and 33 CHPS.

Inclusion/exclusion criteria

Health professionals' clinical practitioners, patients, and patients who have been through the referral system within the municipality were also included in the study. Persons who were not referred within the Hohoe Municipality were not recruited for the study.

Selection criteria

The study included health workers working at the various health facilities within the municipality. Additionally, patients and relatives of individuals who have been through the referral system before within the municipality were included. The study employed a purposive sampling approach in identifying and selecting respondents for the study. Health workers with 5 years of working experience were recruited for the study. Selection of health workers based on the person's experience with referring patients or receiving referred patients. Such health providers identified at the various facilities at the time of data collection were recruited. A total of 9 health workers were interviewed from the Hohoe municipal hospital and one from a CHPS compound at Alavanyo. Seven relatives of patients who had experienced or been through the referral were interviewed. These relatives are considered to have witnessed referral at the health center. Relatives who were found at the health facilities that were caring for their patients and had passed through the referral system were recruited. The study observed patients and health staff during data collection. Also, there was no sample size calculation, so the sample size was determined by saturation. Saturation was ascertained at the point when participants were no longer making new revelations. That is when participants started repeating some insight that had been made from previous interviews by the initial participants. A total of 17 respondents were the study participants. Of the 17, ten were health workers while 7 were patients.

Data analysis

An interview guide (**Supplementary Data 1**) was used as a tool for data collection. In-depth interviews were conducted and tape-recorded with permission from the study participants. All tape-recorded interviews were transcribed into the English language by research assistants competent in the English language. All transcriptions were taken through additional quality checks. A thematic approach was used for the Qualitative software package Atlas. ti. was used for coding and analysis.

Methodological rigor

The study was conducted following the scientific protocols and approved methods and materials. We ensured the reliability of the interview guide, thereby having peer review and external reviews. It was developed through dialogue explorations made by the authors and considering what is known about the phenomenon. Afterwards, the guide was piloted or pretested in the Hohoe municipal hospital to assess how strong the interview guide was in achieving the aim of the study. The piloted interview was transcribed verbatim for coding with Atlas ti. This informed the development of themes and sub-themes, which were developed considering the piloted exploration made. Development of themes and sub-themes was done based on the consensus made by the authors of the study to ensure credibility. After transcribing the actual interviews, the author had deliberations on the pre-developed themes and sub-themes to ensure strong thematic analysis. These influenced the trustworthiness of the study.

Ethical issues

Ethical approval for the study was sought from the University of Health and Allied Sciences-Research Ethics Committee (UHAS-REC), UHAS Ethics Review Committee, and School of Public Health. With Ethical approval number UHAS-REC A.1[41] 21-22 Permission to carry out the research was sought from the Hohoe Municipal Hospital. Respondents were assured of confidentiality, that under no circumstance will their names and other details be linked to the data analysis and dissemination of findings of the study. Data management, storage, analysis, and reporting were done in formats that will not expose the details of respondents.

Possible benefits and risks

There were no direct benefits for participants in this study. However, respondents were informed that the information gathered through your participation in this study will be used as evidence to inform policymakers about the referral system in the Municipality. There are also no direct risks associated with the study, all coronavirus disease 2019 protocols would be observed in order to avoid the spread.

RESULTS

Introduction

The results are presented in three sections: 1) Themes generated on the referral at the CHPS, 2) Themes generated on the referral at the District Hospital, and 3) Relatives' challenges in the referral. In this section the results are presented in order of dominating and consensual themes, followed by minority themes, areas of conflict, and contradictions are presented where these occur in respondent's individual or collective narratives. Quotes that best capture shared or unique ideas are presented for illustration. Respondents were identified by numbers, corresponding to the detailed respondent profile in **Tables 1** and **2**.

Table 1. Demographic characteristics of the respondents

Demographic variables	Total No. of respondents
Sex	
Male	7
Female	3
Ethnic group	
Akan	4
Ewe	3
Ga	1
Others	2
Religion	
Christianity	9
Islam	1
Marital status	
Single	4
Married	6
Occupation	
Medical officer	1
Medical surgeon	1
Gynecologist	1
Nurses	5
Midwife	2
Years of service (yr)	
2	1
3	5
4	2
5	2

Table 2. Socio-demographic characteristics of relatives of referred patients

Demographic variables	Total No. of respondents
Sex	
Male	2
Female	5
Age (yr)	
25–30	2
31–35	1
36–40	1
41–45	1
46–50	0
≥50	2
Marital status	
Married	7
Ethnicity	
Ewe	3
Others	4
Religion	
Christian	3
Islam	4
Education level	
Basic education	3
Secondary	2
Tertiary	2
Occupation	
Teacher	2
Trader	2
Farmer	1
Driver	1
Seamstress	1

Socio-demographic characteristics of healthcare workers

Ten healthcare providers were interviewed in the study. These 7 male healthcare providers and 3 female healthcare providers. Four belonged to the Akan ethnic group, 3 Ewe, 1 Ga, and 2 other ethnic groups. Most of the participants (n = 9) were Christians. More than half (n = 6) were married and the remaining, 4 were unmarried. Most of the participants were nurses (n = 5) and midwives (n = 2). The half of the participants (n = 5) have been in the profession for at least 3 years.

Socio-demographic characteristics of relatives of referred patients

The study also examined the relatives of the person who have been referred before or are currently on referral. A total of 7 participants were interviewed. Most of the participants were females (n = 5) while 2 were males. More than half of the participants were aged (25–45) (n = 5) and 50 years and above (n = 2), none of the participants was aged 46–50 years (n = 0). All the participants were married. Three of the participants (n = 3) belonged to the Ewe ethnic group while the other 4 belong to other ethnic groups (n = 4)

Based on religion 3 of the participants were Christians and the other 4 were Muslims. With their level of education level, 3 had attained some level of education that is the basic level, 2 attained secondary education, and 2 attained tertiary education. Five of the participants were in the informal sector of occupation, while 2 was in the formal sector of occupation.

How referral works within the CHPS

This study discovered that there are three ways by which referral works within the CHPS level (Table 3). These are patients assessment, simple management, and referral of patients.

Patient assessment

Assessment is the first step of referral. When the patient arrives, an assessment is carried out to know the severity of the condition before initiation of medical treatment. In other cases, after the assessment, the patients are either referred based on the capacity of the facility or the capabilities of the health workers. Below is a quote from the respondent at the CHPS on the patient's assessment.

“When the client comes, we assess the client, what we do at our level here is simple malaria i.e., uncomplicated malaria, if the person is having fever, we try to do the tepid sponging.” (Health worker 1)

Simple management and drug administration

The second way referrals operate in the CHPS is simple management and drug administration. During this phase, management practices like tepid sponging and drugs are administered before patients are discharged. However, patients are referred when there are complications beyond the management of the facility. This is how the midwife at the CHPS compound highlighted how cases are managed and referred,

“Start with paracetamol depending on the age, then after that we detain for some time. We do tepid sponging then we check after some time we check again then if the temperature goes down accept the client if it is uncomplicated malaria we can give tablet we give then we make sure the person goes to the house and continues but if it is beyond that i.e., we try to normalize the temperature or reduce it but if we realize the

Table 3. Thematic table

Main theme	Sub theme	Codes
Referral at CHPS	Cost of ambulance	Means of transport The outcome of unaffordable patient Ever tried using the ambulance Ever referred patient Frequency of referral Challenges Visit referred facility Results from referral Interpretation of results Conditions referred Pre-information of the referred facility Documents sent to the referred facility Follow up of referral Absence of a health staff Recommendation Enough health workers
	District hospital	How referral works Criteria for referral Enough resources Feedback Mode of transport Recommendation State of the patient before referral Worst situation encountered
	Relatives	Accompanied by health staff Situation before referral Recommendation Challenges Arrival Given referral letter

CHPS = Community-based Health Planning and Services.

condition is worsening or malaria makes the person unconscious the only thing, we stabilize the person we call taxi.” (Health worker 1)

Referral of patients

Referral of patients and means of transport was the final stage noted on how referral works at the CHPS level. Generally, there were no ambulance services; however, they use taxis as a means of transporting patients. It was also noted that the referrals were done on the patient's preference. The patients and care takers decide where they go for continuity of seeking health care services. This is what the midwife at the CHPS compound said.

“We have two stand-by for us, so we just call one or we near the roadside so sometimes we just ask a relative to get a taxi some of them prefer to go to Kpando some also prefer to Hohoe hospital. So we getting transportation is not a challenge to us at all.” (Health worker 1)

Attempts to access ambulance services

Though the ambulance services are at the district health facility. Participants mentioned that the ambulance has never been used within the CHPS compound. This is because the facility has not recorded any severe cases that required ambulance services. It was however noted that the ambulance number is available for emergency cases. Though the CHPS has never used its services. This is a coat to illustrate ambulance utilization at the CHPS level.

“We haven’t tried using it before we always have readily available means of transport and we haven’t had any case that had become very critical that we have to say we rushing someone to the next facility that kind of emergencies comes with maternity issues but we just opened a maternity unit so we haven’t had any case as such but we have the ambulance number if there is any case that a need for an emergency for referring, we do that but we haven’t called any yet.” (Health worker 1)

Cost of ambulance services

Regarding the cost of ambulance services, the midwife at the CHPS compound made it known that the services are not free. It comes at a cost for fuel maintenance and drivers’ stipends. Before the patient is been transported for further management, they must pay this cost.

“No they take money. We haven’t called an ambulance in the CHPS compound here but in the hospital setting where I worked, mostly call them the ambulance takes some monies for fuel from the clients before they are transported for further management.” (Health worker 1)

The study found that patients who are not able to afford ambulance services are denied services. This puts the health workers at the CHPS compound to work extra even when they know it’s beyond them. However, the respondent at the CHPS highlighted that such incidents rarely occur at her facility and that she has never witnessed any. Below is a statement by the Midwife at the CHPS.

“In all the cases I have witnessed they try and arrange to get those monies. You know what would happen the patient would just die or keep doing what we were doing to revive the client.” (Health worker 1)

Conditions mostly referred at the CHPS

At the CHPS level, only limited and basic services are provided. However, pregnancy complications and high blood pressure were the cases mostly referred to at the CHPS level. Though the CHPS does not manage complicated conditions, referring them has been the best option. But in the case of this study, it was noted that only hypertension and pregnancy-related complications were the most preferred. It was also noted that aside from these two conditions, the facility hardly gets conditions that may require referrals. Below is a statement from a midwife at the CHPS compound.

“No we haven’t referred any other cases, those cases we refer the high blood pressure but they come we check their BP, and if it is high the following day we ask them to come because you can’t use a day to conclude so they come like a week or 3 days and we tell them it has risen so we just write sometimes you have to go to the hospital.” (Health worker 1)

Communication with referred facility

The participant said referral letters were not used at the CHPS level. Instead, they reach the next facility by a phone call informing them about the referred patient. They also aid in diagnosis by informing the facility about the suspected condition the patient might be suffering from. In some instances, the patient would just be asked to go to the next facility on their own and join the queue to assess further management. The quote below illustrates the response of the midwife at the CHPS compound.

“So they just go and follow the queue. Its only one occasion when a pregnant woman came in with ‘vaginal warts’ so she had to be referred so with that one I had to refer her to ANC in Hohoe so I had to pre-inform a midwife there because they are my colleagues and she had to go there with her means and without a referral letter.” (Health worker 1)

Documents sent to the referred facility

Participants emphasized that patients who have been referred are not given any referral letters instead, they report to the hospital with their ANC cards, and the midwife documents in the card that the patient needs these services and has been asked to go.

“They go with their ANC cards sometimes I write in their cards that this client needs to have a hospital delivery when it happens and you come to antenatal at my end, I tell them that they have to start going to the hospital so that when their time is due, they are just captured over there at one so they communicate with us that we should be doing follow up on the clients. So that’s how we communicate.” (Health worker 1)

Interpretation of results

The study discovered that the hospital reports and outcomes of the treatment are always done. Even though the health staff gives these documents to patients, they don’t get time to interpret the results for them. However, most of them also see other health professionals interpret it for them. Below is the statement by the respondent at the CHPS compound.

“Because of time some when they go, they get there as and if everything is done for them, they pass through the ANC and ask a midwife to interpret it for them some to say is getting late so they take the result and bring it to me to interpret.” (Health worker 1)

Recommendation

The respondents at the CHPS level recommended the need to improve referrals. There was a need to increase the human resources at the facility to enhance the effective running of the CHPS.

“Well, I would say the means for transportation is not a problem relating to my facility but the only challenge is with my clients, the only thing is to have a durbar and educate them on so many conditions and that would make them have adequate knowledge on the conditions so that we would not have to talk and talk and even bringing people on to explain. Because if u say this time you need to refer them. Is even surprising that people are ready to support them by transporting them but the clients themselves say it’s not necessary in short there is reluctance to go.” (Health worker 1)

Referral at district hospital

This section involved (n = 9) respondents from Hohoe municipal hospital. An in-depth interview was conducted on how referrals work, challenges, and possible recommendations on the preferred referral system.

How referral works within the district hospital

Respondents explained the processes involved in referrals. The first process is to pre-inform the next facility to check for space and the capacity of the facility in handling such cases. After this, arrangements are made for an ambulance before they refer.

“The Doctor would have to come and review and request for the referral before we process it into a book then we call the referral center to inform them about the condition. We call the ambulance as well and the paramedics make themselves available as well.” (Health worker 9)

Some respondents described the referral process as; referring cases beyond their capacity and needing specialist care or further management after diagnoses are made. They refer to tertiary or quaternary health facilities to continue management.

The comments below illustrate how respondents alluded to the notion:

“Referral is all about cases that need special care or that are above the doings of the hospital. Things that maybe we are not able to handle and needs a specialist that’s when we refer and most of our referral centers is the Ho teaching hospital and Korle Bu sometimes we refer to saint Joseph orthopedic hospital.” (Health worker 8)

“Depending on the case if they don’t have a specialist to manage that particular diagnosis, we refer especially orthopedics there no surgeon here for orthopedic cases so we refer, neurological cases we refer also some also would want to go to the referring point because of proximity to the family so with that we refer, we do refer kidney cases those that need dialysis, liver conditions and CVA with neurological deficit.” (Health worker 7)

Referrals coming from the lower facilities were mentioned. According to a respondent 6 Referrals come from the CHPS and the patient is with a referral letter indicating how management commenced.

“With the referral system we have from the CHPS coming into the municipal hospital with that one patient come with referral letters indicating how management was commenced from the lower facility to the hospital and with that we barely have health workers accompanying the case to the facility but from the municipal hospital or other district hospitals they usually call the hospital to arrange for the referral but they are cases where calls are not been placed at all but we end up finding patients here but if we want to refer from here to the other facilities we normally.” (Health worker 6)

Criteria for referral

majority of the participants highlighted the patients who needed specialist care or further management are mostly the criteria that guide them to make referrals. Here are some of their responses.

“Either we don’t have the experts or the equipment to manage it at your facility. Maybe a lady comes here with dilated cardiomyopathy something that would need an (...) or the case would eventually need intensive care unit (ICU) care or the baby is too small for us to manage at our ICU so in that kind of case we weigh it after the mother has given the baby to us than in that case, we say you can go to how because you can get ICU care.” (Health worker 2)

“Severe head injury as I said am with the surgical team so most the case we refer like compound fractures, fractures which we can’t manage here but most of the RTAs the minor ones we can manage here but if it involves the bone, we have to refer you I don’t

know if the medical cases because am not with them but for the surgical cases is just the wild one's open bone fractures those need a specialist that one, we just refer them.”
(Health worker 4)

Health workers' challenges

This study explored the health worker challenges that affect the referral system. These challenges were categorized and thematized in order of dominance.

1) Human resource challenge

The first challenge was human resource. Four of the respondents noted that the referral system lacks enough human resources. This stifles the smooth running of the referral system. For instance, because there are not enough human resources, during referral, patient cases deteriorate before they get to the next point of referral.

2) Medical equipment challenge

Most of the respondents highlighted that the district hospital lacks some medical equipment. Some of this equipment exist but have been faulty and has not been repaired. This affects most of the referrals as most patients referred from CHPS, health centers, and other facilities may not get the needed health care services.

“It's a challenge at times the ambulance comes in and they don't have an oxygen and assuming the patient needs oxygen we would have to give our cylinders out to them.”
(Health worker 3)

3) Ambulance services

This study found that the district hospital's ambulances were not enough. This affects referral systems as people who require urgent transportation may not get it. The existing ambulances do not have oxygen in them, and this made it difficult to transport patients who may require oxygen on their way to the next facility.

“Sometimes oxygen the ambulance coming is not having oxygen in but the patient needs oxygen to go so we have to give out our cylinders they would so, be sent and later brought.” (Health worker 6)

Feedback

Most respondents pointed out that there has not been any form of feedback mechanism to the extent some haven't even seen any of the feedback forms and instead they call if they are interested in the case. only a few spoke about receiving feedback from other referral points. Here are some of their responses.

“The feedback I know there would be feedback but I have not seen any before there should be.” (Health worker 8)

“We don't get feedback sometimes when we are interested in the case we call and ask how the patient is doing but that's once in a while.” (Health worker 5)

Recommendations made at the district hospital

The health workers recommended the following for dominance to be put in place for a good referral to be in place. More ambulances and a fund for fueling the ambulances, a working

base, and a knowledge base have to be expanded, communication improved, and a good road network.

More ambulances and a fund for fueling the ambulance

Three participants recommended that for an effective referral to be in place, there would have to be more Ambulances in the district so that if one has sent a patient to another health facility, there would be a readily available one to serve another patient. And a fund to help cushion the amount the patients are expected to pay before referral, even though it is supposed to be free.

“The ambulance they can call an ambulance and they would tell you they don’t have an ambulance in the region the one we have is broken down. If those things are in place, I don’t think referral should be a challenge.” (Health worker 8)

“We need more ambulances, a fund in place may be ambulance fund so that those who can’t afford would at least get some money to cushion them to the next point of referral.” (Health worker 7)

Working base and knowledge base improved

A participant highlighted that the working and knowledge base needs to be expanded to be able to manage some cases before referring for continuity of care because ideally, their work base is not enough to be able to manage cases down here at the district hospital.

“The work base and knowledge base should be expanded so that most cases would be managed in the middle-class facilities like the district facilities and teaching hospitals.” (Health worker 8)

Improved communication

The health workers also mentioned Improved communication as a way the referral system could be improved. where they are aware of a patient coming and necessary preparations made.

“Telemedicine where calls are made have been introduced if the government has instituted the telemedicine where calls are made to the nest facility. You would be there and a CHPS compound would refer a case without even your notice. I mean you would have to get oxygen onboard and other stuff also on board so if it’s worked on the communication system would be intensified.” (Health worker 3)

Good road network

The good road network was also mentioned as an area that needed urgent attention because patients referred in or out of the district either had their conditions deteriorate or lost their lives because of the bad nature of the roads, despite the availability of the Ambulance service.

“The road network should be worked on. Most at the times the ambulances are available it is in rare cases that they aren’t available but the road networks should be worked.” (Health worker 9)

Challenges faced by relatives of persons on referral

The aspect of the study explored the challenges relatives go through during referral. They were categorized and thematized in the order of dominance.

Bad roads

Participants mentioned bad roads as one of the most pressing challenges they encountered moving to the referred facility. They noted that bad roads contributed to worsening patients' conditions. Talking about this, a relative alluded to the notion.

"By grace, we came successfully by taxi, but the bad nature of the road was a challenge because he had a problem with his prostrate." (R1)

Financial challenges

Financial hardship was mentioned or indicated by participants as an impediment to transferring patients to the next referral health facility. Other participants said that, although an ambulance was available, they could not afford the transportation fee. The quote below shows how she put it.

"Getting a car was a bit challenging ambulance was there but we couldn't afford." (R2)

The study also found that there were not enough wheelchairs or stretchers to convey patients in critical conditions into the ward, also health workers were reluctant to attend to patients even in emergency settings.

"On our arrival there was no wheelchair to pick him into the ward there was no nurse to assist us until I meet a man in a white coat and he called two nurses to come to our aid." (R7)

The study also revealed that there were not enough beds, so some patients were attended to while sitting on a chair.

"There was no bed and they asked us to wait for a while my grandmother was sitting and been attended to so all that was done before the bed was made available." (R1)

Accompanied by a health staff

Six out of the respondents went on the referral without any health staff accompanying them. Only one of them was able to afford to come with an ambulance, so paramedics were on board until they found themselves at the hospital.

Recommendation from the relatives

Some of the relatives of patients who have been on a referral made their suggestions on what should be done to improve the referral system, except for 3 participants who felt some of the recommendations are as follows: good communication amongst health facilities, health workers being committed to their duty, and there should be readily available of oxygen.

Good communication amongst health facilities

A participant considered the communication between the referral points very bad. Ideally, there should be good communication between the referral points for an early continuation of treatment or further management. The fourth relative recounts her experience as a recommendation.

"I think it shouldn't have been something that would be difficult to refer us to Tamale. Maybe they had a misunderstanding amongst themselves otherwise they wouldn't have

rejected us. Because wanted to be closer to our relatives at Tamale. Once a doctor sees a signature on the referral letter, I don't think it should be a difficult task.” (R4)

Commitment of health workers to their work

Most of the respondents recommended that the health staff should be committed to their work.

“As I run to them in this emergency ward, I need a wheelchair they should have gotten up to get us one but no one did that. Even though we are in the era of disease they should have sanitized their hands or put on gloves to help bring my dad out of the vehicle look at me a woman I was suffering.” (R4)

DISCUSSION

The findings from this study showed that Health workers had challenges concerning human resources and medical equipment. There was the unavailability of a specialist to manage certain cases like severe head injuries or cases of Road traffic accidents, so instead, patients' cases deteriorate or die even before they get the next point of referral. Also, there were bad road networks and the unavailability of enough Ambulances within the Municipality. This is in line with Research conducted by Ghana Statistical Service¹³ that found shortage of skilled health care workers, equipment, and drugs is a factor that affects the provision of quality referral care at the primary health care level. This leads to self-referral and bypassing the primary health care facilities. Also, research conducted by Chitsa¹⁴ found that insufficient health experts, inadequate infrastructure supplies such as limited bed availability, and inadequate drugs and blood banking services are other threatening challenges to the referral system.

The study revealed that there was no feedback mechanism, which affects the tracking and feedback on a patient's condition. Bad communication between the health workers was also discovered in the study because receiving facilities do not receive any prior information on the arrival of the patient; there would have to be adequate preparations made before the patient arrives. Similar research conducted in the Obstetric Unit of the Obstetric and Gynecological Department of the Korle Bu teaching hospital found that there was a poor feedback system whereby receiving facilities do not receive prior information upon the arrival of a patient, and no feedback is sent to the referring facilities after the patient has been treated. This is one of the challenges that affects the referral system. This act makes it difficult for health facilities to properly assess themselves.¹⁴ The same research also found out that there was a lack of adequate information on the referral letters, so they would have to go through examination then come out with the diagnosis, also some health practitioners do not manage the patients well before referring them so they intend to hide the information from the receiving facilities which make it difficult in continuing the treatment of the patient.

The study revealed some challenges relatives experience during referral. Some of these were financial challenges, bad road networks, unavailability of oxygen, and some health workers not being committed to their work. Similar research conducted by Give et al.⁷ found that the distance to the health care facilities, lack of access, and cost of transportation are major barriers that prevent patients from following the referral paths.

Also, Daniels and Abuosi¹⁵ found out that socioeconomic or cultural factors such as poverty, taboos, and beliefs are potential causes of delay in the decision to seek care. However, physical

accessibility factors can be a reason for non-compliance of patients to follow the referral. Many low-resource areas have poor roads, bad weather conditions, and lack enough health facilities and health experts, which makes it difficult to go for further treatment. In such instances, they rather seek care at home from traditional healers and traditional birth attendants rather than seeking care at more appropriate and sophisticated care referral facilities.

Again, qualitative research conducted in the Upper West Region of Ghana found that road networks and transportation are identified as determinants of poverty. It becomes a dire situation when rural dwellers with low income are challenged with the availability and high cost of transport when seeking health care. It was further identified that transporting a patient on a bicycle is the most common means (excluding women in labor) of transporting people to the CHPS compound and beyond. It also found that some have to go on foot or be carried at the back by two or more people, and this takes them a lot of time to cover a short distance. It was also discovered that some would have to wait for market days when there would be market cars passing through these remote areas. Whereas a market day is once every six days, it implies that the patient would have to wait for five days before having a means of transport service to access health care services. Inaccessibility of transport services by the community these days influences them to rely on local medicine to manage the illness until the market day is due. The same study reported that there are free maternal health services that are run alongside the CHPS in rural areas; however, the absence of transportation is harming the potential impact of the program.¹⁰

In conclusion, referral systems and pathways are essential in the various levels of decentralization of the health care system, especially the CHPS. Generally, this study unveiled that the referral system has many challenges that impede its proper operation. This study found that the referral at CHPS faced several challenges, such as bad roads, no referral letters, and ambulance issues.

The study also found that the challenges of the referral system at the CHPS compound were the same as those of the district hospital. Human resources and faulty medical equipment were also noted at the district hospital level. The relatives of patients who have been referred or are on referral also noted some challenges, such as the commitment of health workers to the patients, bad roads, and the cost of referral, especially with the means of transport.

Based on the findings from this study, the following are therefore recommended to make the referring systems, both at the CHPS and district health facility levels, more effective.

The Government of Ghana should consider investing in the Ghana Ambulance Service for improvement in terms of communication and feedback. Instead of paper mode of communication and feedback, digital platforms should be integrated in the referral system to ensure an easy flow of information between health facilities.

Also, governments should subsidize the cost of ambulance services through the national health insurance policy to make the cost of service bearable, and there should also be flexible payment options for those who may not have money at the time of referral.

The Hohoe municipal assembly should ensure the construction of roads and ensure regular maintenance on the roads to prevent delays that may worsen the patient's condition before they reach the point of treatment.

Finally, the municipal directorate should organize more human resource training on the management of complications to strengthen the capacity of healthcare providers. The CHPS should be empowered to manage more complex cases with improved facilities and human resources to reduce the burden on the district health facilities.

The Government should provide allowances that will serve as motivation for care providers, especially, those in the CHPS compound. This will serve as a drive for care providers to give it their all in saving patients.

There were a few limitations to this study. Participants were not able to express themselves explicitly because the discussion or interviews were not done in the participants' natural settings, and this made the participants unable to open up about their challenges in the referral system. Moreover, given the qualitative nature of the study, generalization is not assured. This has affected the ability to make reference to other part of the country given the number of participants. Although the same system exists in other regions, yet the explored challenges as revealed by the health workers and care givers might differ in other regions. However, the exploration made portray a picture that indeed the referral system is battling with challenges in other regions. Despite these limitations, the study is relevant and contributes value to the body of literature that has solely focused on only a few parts of the country, but with similar referral challenges like Hohoe.

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SUPPLEMENTARY MATERIAL

Supplementary Data 1

Interview guide

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